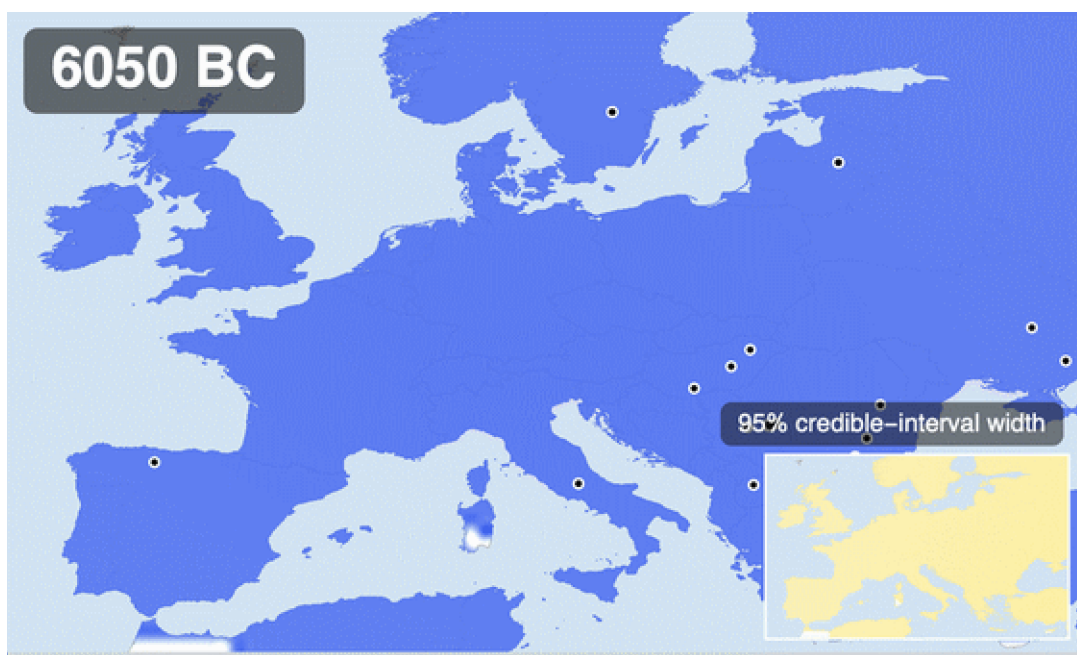


Watching evolution happen: lactase persistence in 8,000 years of ancient DNA

From raw ancient genotypes to a spatial diffusion–selection model, fitted with sequential Monte Carlo ABC — with the uncertainty carried all the way into the pictures



terior mean frequency of the lactase-persistence allele (rs4988235–A) across Europe over the last 8,000 years (about 6050 BC to A 1950), on the TemperatureMap scale. White-ringed dots are the ancient individuals dated within ± 400 years of the moving window; the inset tracks the 95% credible-interval width — where the map is honest about not knowing. This copy is the output of the ExportHeroAnimation call executed in §12 of this very notebook; it is placed up front so the destination is visible before the journey.

Around nine thousand years ago, everyone in Europe — every single adult — lost the ability to digest milk sugar a few years after weaning, just as almost all mammals do. Today, most northern Europeans can drink a glass of milk without a second thought. The switch that changed, a single letter of DNA about 14,000 base pairs upstream of the lactase gene, is the strongest signal of recent natural selection anywhere in the human genome — and thanks to ancient DNA we no longer have to infer its history from modern variation: we can simply watch it happen, skeleton by dated skeleton, across eight millennia.

Two recent results make this more interesting than the textbook just-so story. **Evershed et al. (2022, Nature)** assembled the largest set of prehistoric milk-fat residues from pottery and showed that Europeans were dairying enthusiastically for thousands of years while the allele stayed rare — and that a selection model tracking milk use explains the allele's trajectory *no better* than plain uniform selection since the Neolithic. **Burger et al. (2020, Current Biology)** showed the flip side:

from a Bronze Age battlefield frequency of about 7%, reaching today's values requires a selection coefficient of roughly 0.06 per generation over the last 3,000 years — enormous by the standards of human evolution. Milk was everywhere, the allele was nowhere, and then, quite suddenly in evolutionary terms, it was everywhere too. Why?

This post does not settle that debate. It does something more modest and, I hope, more reusable: it builds the *entire* analysis pipeline in Wolfram Language, from the public ancient-genotype workbook to a calibrated spatial model — and treats uncertainty as a first-class output at every step. We will parse and clean 2,999 ancient genotype records, reproduce the classic regional frequency curves with proper binomial likelihoods, build a spatial diffusion–selection model on a landmasked grid of Europe, fit it with sequential Monte Carlo approximate Bayesian computation (10,000 forward simulations), check it with posterior predictive simulation and two kinds of held-out validation, stress-test the priors, and render the posterior — mean *and* credible-interval width — as the animation above. Every figure sits directly below the code that makes it, and the whole thing reruns from two wolframscript commands (§13).

All the heavy lifting lives in the companion package `LactasePersistenceSpatial.wl` — the only file you need beside this notebook (plus an internet connection for the first data download). Put the two files in one folder and evaluate: the setup cell finds the package next to the notebook (or in `src/` when you are inside the cloned repository), the data download and every output recreate themselves relative to that folder, and the expensive SMC posterior reloads from stored result tables when present and refits from scratch when not. Every output you see below was produced by running this notebook top to bottom with wolframscript.

1. Setup and data

Two data sources, one loader. The historical source is the public GLAD ancient-genotype workbook (Global Lactase persistence Association Database), derived from the Allen Ancient DNA Resource v44.3 and used for the Evershed et al. (2022) analysis; `RetrieveRawData` downloads it once with a SHA-256 checksum and provenance manifest, and `WriteProcessedData` parses it into tidy CSVs. The *active* dataset is newer: rs4988235 extracted for every individual in the current AADR v66.p1 release (see `scripts/extract_aadr_v66_rs4988235.py` and `scripts/ingest_aadr_v66.wls` in the repository), deduplicated to one record per individual and validated against the GLAD genotypes where the two overlap (796 of 797 strict homozygotes concordant). `LoadProcessedSamples` prefers the v66 file automatically whenever it is present — including the copy that ships beside this notebook — so the counts below show both generations of the data:

```
In[ ]:= nbdir = NotebookDirectory[];
(* LactasePersistenceSpatial.wl ships next to this notebook; inside the
   full repository the src/ copy and the repo data layout are used *)
root = If[FileExistsQ[FileNameJoin[{ParentDirectory[nbdir], "src", "LactasePersistenceSpatial.wl"}]],
  ParentDirectory[nbdir], nbdir];
SetDirectory[root];
Get[First[Select[
  {FileNameJoin[{nbdir, "LactasePersistenceSpatial.wl"}],
   FileNameJoin[{root, "src", "LactasePersistenceSpatial.wl"}]}, FileExistsQ]]];
```

```
In[ ]:= raw = RetrieveRawData[Directory[]];
processed = WriteProcessedData[Directory[], raw];
samples = LoadProcessedSamples[Directory[]];
<|"GLAD (AADR v44.3) called samples" -> Length[processed["CalledSamples"]],
"active dataset (AADR v66.p1)" -> Length[samples]>

Out[ ]:=
<| GLAD (AADR v44.3) called samples -> 1785, active dataset (AADR v66.p1) -> 10119 |>
```

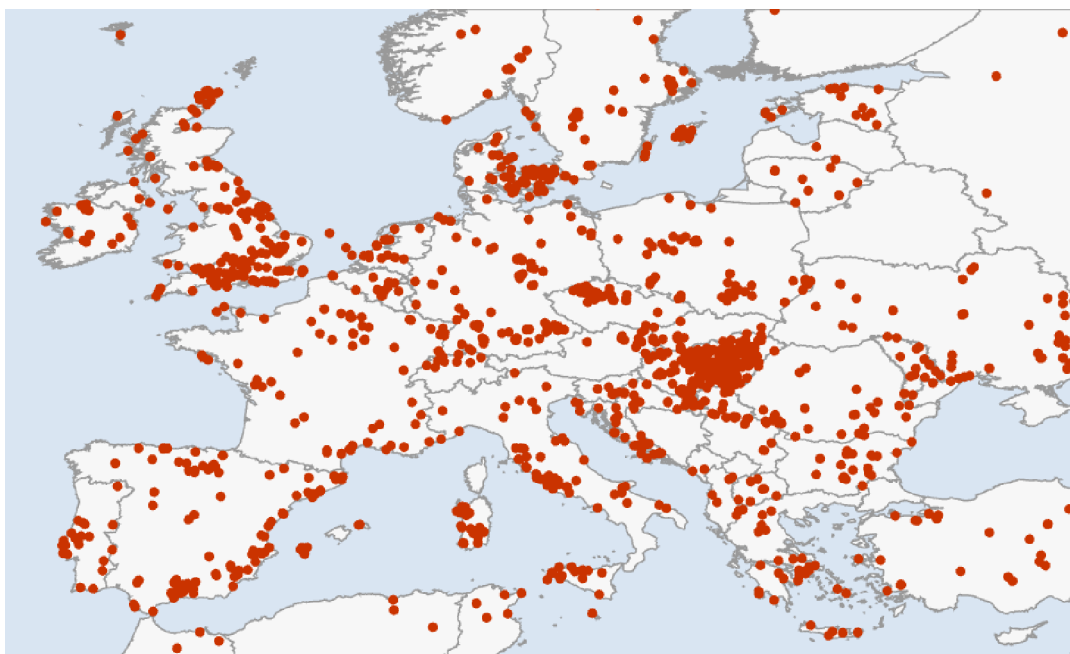
Parsing ancient genotypes is where silent errors live, so the cleaning rules are explicit and tested: genotype strings are normalized so that A — and the strand-flipped T, which occurs in exactly 3 heterozygous calls — count as the derived (persistence) allele and G/C as ancestral; a single-letter pseudo-haploid call contributes one called allele and a diploid call two; coordinates accidentally recorded in millidegrees (37724 for Sicily) are rescaled; and every sample is assigned to one of four coarse analysis regions echoing the published framing. Of 2,999 rows, 10119 have a genotype call plus usable age and location.

```
In[ ]:= Counts[#, {"Region"}] & /@ samples]

Out[ ]:=
<| Rhine–Danube -> 3370, Mediterranean -> 1437,
Outside Europe -> 3865, British Isles -> 803, Other Europe -> 230, Baltic -> 414 |>

In[ ]:= GeoListPlot[GeoPosition[{{#, "Latitude"}, {#, "Longitude"}}] & /@ samples],
GeoRange -> {{34, 63}, {-12, 36}}, GeoProjection -> "Equirectangular",
PlotStyle -> Directive[Opacity[0.45], RGBColor[0.153, 0.51, 0.64], PointSize[0.005]],
GeoBackground -> GeoStyling[{"CountryBorders", "Land" -> GrayLevel[0.97], "Ocean" -> RGBColor[0.8
ImageSize -> 620]

Out[ ]:=
```

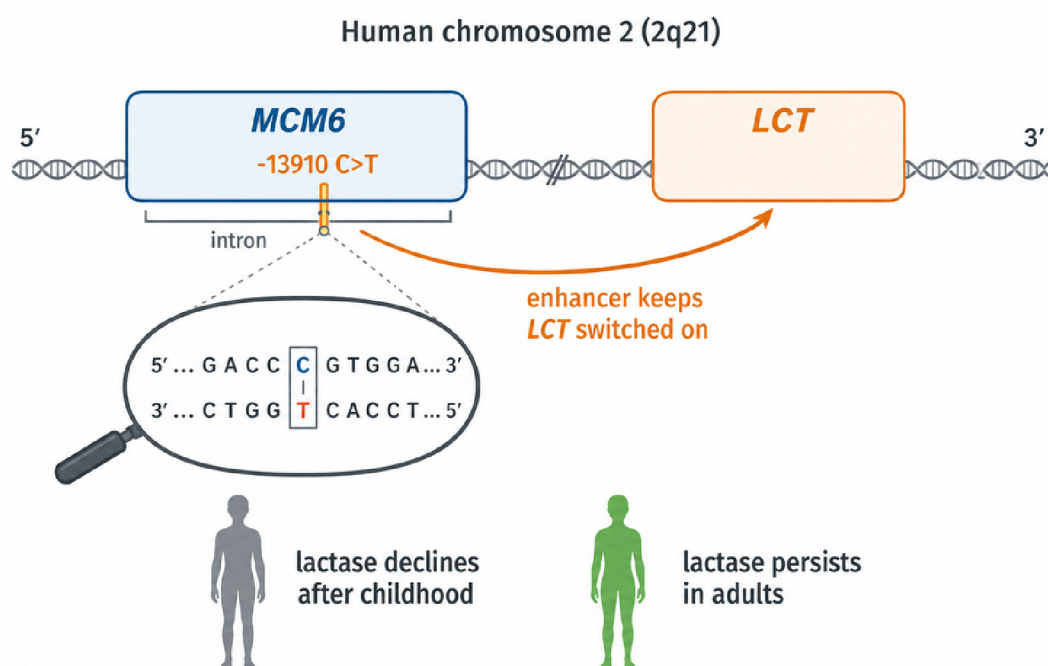


Where the 10119 usable ancient individuals come from. Density is wildly uneven in space and time — the Mediterranean is rich, the Baltic sparse, and almost nothing is younger than 2,500 years — and that unevenness is exactly why the model must carry its uncertainty around explicitly.

2. What is lactase persistence?

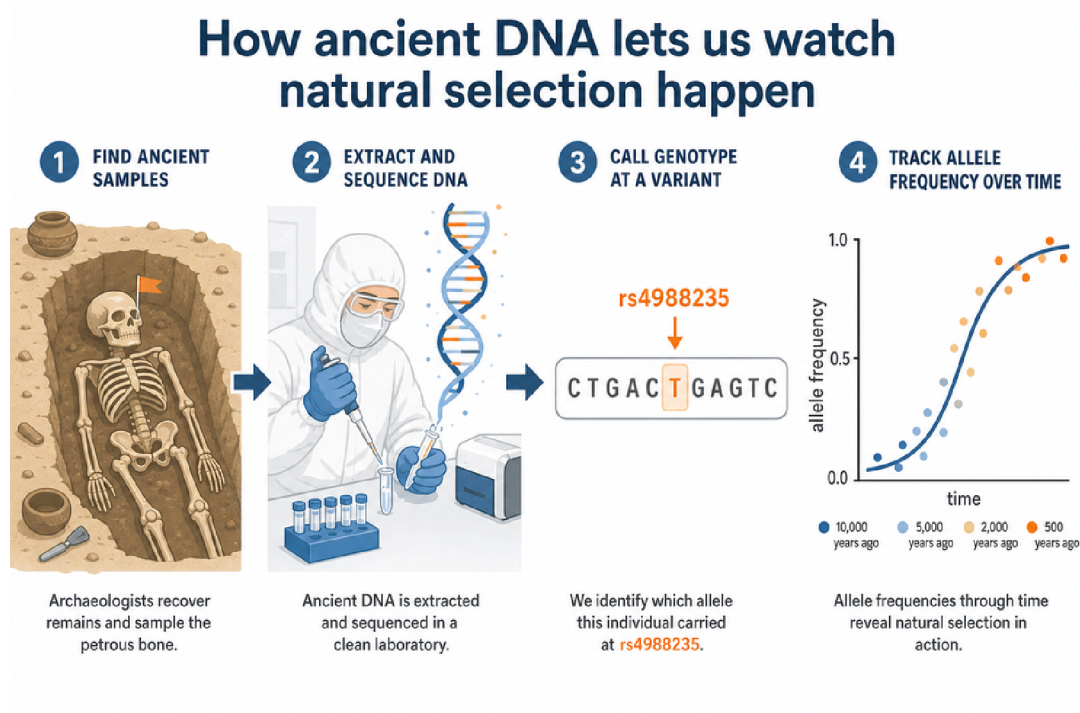
Lactose, the sugar in milk, cannot be absorbed directly: it must first be split into glucose and galactose by the enzyme *lactase*, produced in the lining of the small intestine and encoded by the *LCT* gene. In the ancestral mammalian program, lactase production is switched off after weaning — an adult wolf, cow, or (ancestrally) human who drinks milk gets bloating, cramps, and worse, because undigested lactose ferments in the colon. The European persistence variant is not a change to lactase itself: it is a single C→T substitution 13,910 bases upstream, inside an intron of the neighbouring gene *MCM6*, that acts as an enhancer and simply keeps the *LCT* switch turned on into adulthood. On the reported strand of the association database this is the G→A change at [rs4988235](#) — one letter, one glass of milk, and (as we will see) several percent of extra surviving children per generation at its peak.

External asset. The gene schematic was generated outside Wolfram Language with OpenAI's gpt-image-2 model; the prompts and the one-off Python driver are checked into docs/images/figures-generated/ of the project repository. It is a static illustration; there is no WL code to run for this figure.



A crucial subtlety for what follows: the variant is effectively *dominant* — one copy is enough to digest milk — and it rides on a long shared haplotype, which is how it was first detected from modern data alone. But modern data compress history: they can say the allele rose fast, not *when*. Ancient DNA turns the same question into direct observation.

External asset. The ancient-DNA workflow illustration was generated outside Wolfram Language with OpenAI's gpt-image-2 model; the prompts and the one-off Python driver are checked into docs/images/figures-generated/ of the project repository. It is a static illustration; there is no WL code to run for this figure.



3. The regional story: four binomial logistic fits

Before any spatial machinery, the honest first step is to reproduce the standard non-spatial description: for each region, a logistic trajectory $\text{logit } p(t) = \alpha + \beta (10000 - t)/1000$ fitted by maximizing the exact binomial log-likelihood over every called allele — no time-binning enters the fit; the bins you see in the figure are display only, each carrying a 95% Wilson score interval and sized by its allele count. Standard errors come from the numerical Hessian at the optimum, and the fitter flags any solution that lands on a parameter bound. That flag is not decoration: an early version of this project silently reported a boundary solution for the British Isles, and the only symptom was a suspiciously round α . Fits you cannot audit are fits you cannot trust.


```

In[ ]:= fits = FitAllRegionalLogistics[samples];
Grid[
Prepend[
Table[With[{f = SelectFirst[fits, #["Region"] === r &]},
{r, f["SampleCount"], f["CalledAlleles"], f["DerivedAlleles"],
Row[{NumberForm[f["Alpha"], {5, 2}], " ± ", NumberForm[f["AlphaSE"], {4, 2}]}],
Row[{NumberForm[f["BetaPerKyrTowardPresent"], {4, 2}], " ± ", NumberForm[f["BetaSE"], {4, 2}]}],
NumberForm[f["SelectionPerGenerationApprox"], {4, 3}]}],
{r, MajorRegions[]}],
Style[#, Bold] & /@ {"region", "samples", "alleles", "derived", "α", "β per kyr", "≈s per gen"}],
Frame -> All, FrameStyle -> GrayLevel[0.8], Spacings -> {1.2, 0.6},
Background -> {None, {GrayLevel[0.93], None}}]

```

Out[]:=

region	samples	alleles	derived	α	β per kyr	$\approx s$ per gen
British Isles	803	1050.	410.	-8.72 ± 0.63	1.07 ± 0.08	0.030
Rhine–Danube	3370	4269.	705.	-10.64 ± 0.64	1.09 ± 0.07	0.031
Mediterranean	1437	1768.	94.	-7.87 ± 0.78	0.67 ± 0.10	0.019
Baltic	414	768.	247.	-10.14 ± 1.03	1.16 ± 0.12	0.033

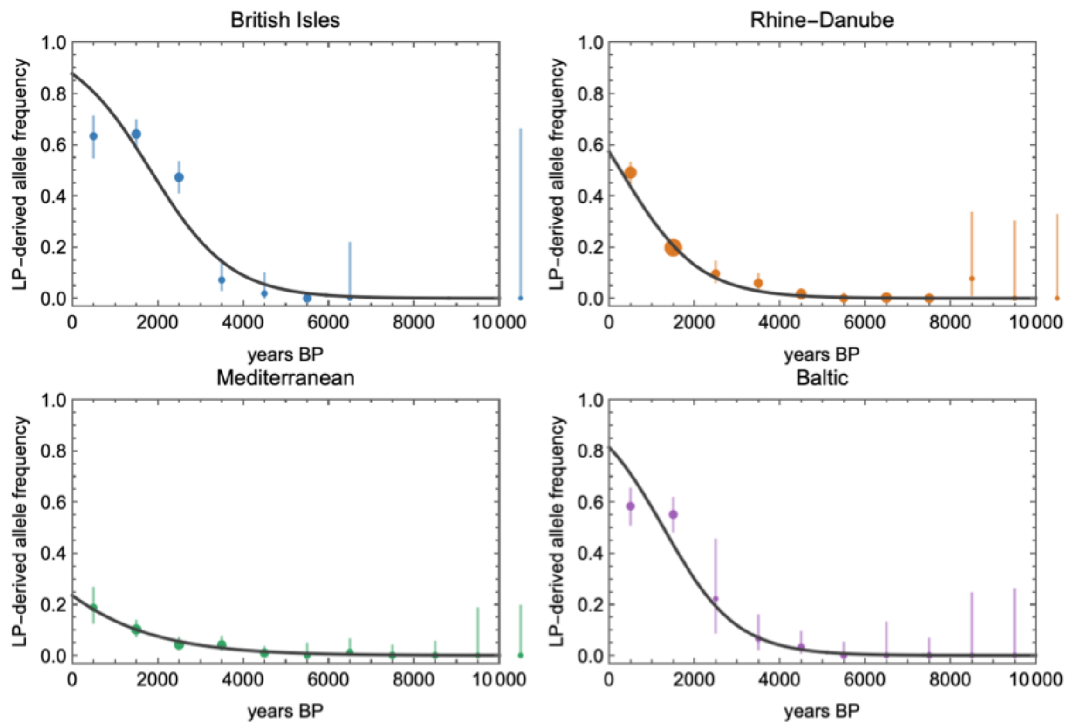
With a 28-year generation time, a slope β on the logit scale translates to an approximate per-generation (genic, codominant) selection coefficient $s \approx 28\beta/1000$. The four regions land between 0.019 and 0.033 — the familiar few-percent picture, strongest in the north-west. Burger et al.'s 0.06 is estimated over the last three millennia only, while these slopes average over eight, so it should and does sit at or above the top of this range: time-averaging dilutes a late, strong pulse.

```

In[ ]:= regionalOutputs = ExportRegionalFitOutputs[Directory[], samples, fits];
Import[regionalOutputs["RegionalFigure"]]

```

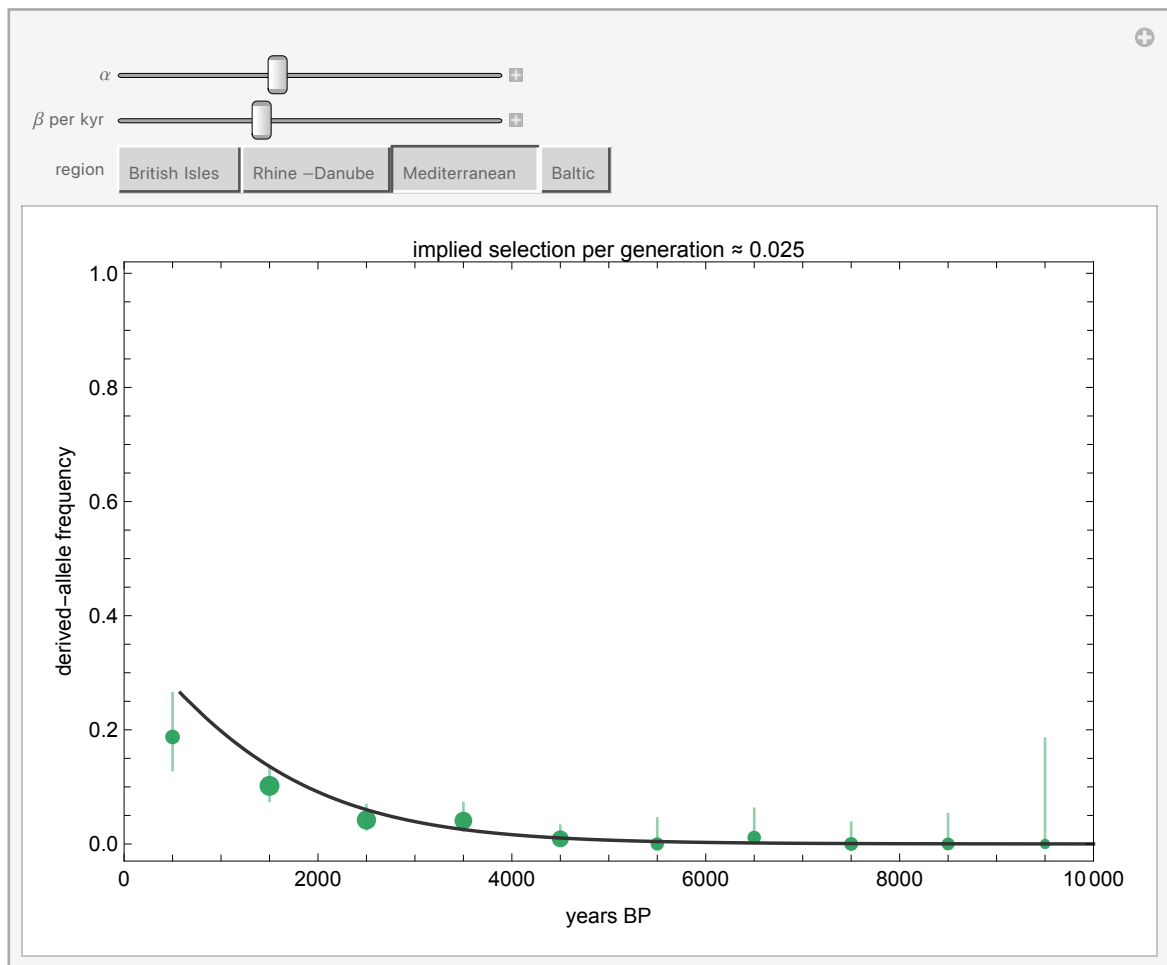
Out[]:=



Observed binned frequencies (Wilson 95% intervals, point area $\propto$ allele count) with the fitted logistic curves. Note the Mediterranean's leisurely rise against the British and Baltic sprints — and note also how far right of the last data point every extrapolation to the present runs. Everything after $\sim 2,000$ BP is extrapolation, and the spatial model below inherits that caveat.

Rather than take my word for how sensitive these curves are, drag them yourself. The Manipulate below overlays an adjustable logistic on each region's binned data — the label converts your β into an implied selection coefficient as you drag. It takes about ten seconds of play to internalise two things: how narrow the corridor of plausible β is for the data-rich Mediterranean, and how wide it is for the Baltic.

In[] := LogisticExplorer[samples]
Out[] :=



4. A spatial diffusion–selection model

Regions are administrative conveniences; alleles do not respect them. The spatial model discretizes Europe (35°N – 63°N , 12°W – 35°E) into a 2° grid, keeps only cells whose centres are on land (191 of them — so Britain stays rook-connected to the continent across the Dover cells, while Ireland becomes an island in the model's adjacency too, a caveat worth knowing), and lets the local allele frequency evolve in 250-year steps from 10,000 BP:

```

p[i, t+dt] = p[i, t] + g ( (s0 + sDairy D[i, t] m[region i]) p (1 - p)
                        + mig mean[ p[j, t] - p[i, t], j adjacent to i ] )

g = dt / 28 generations per step

```

Selection has a baseline component s_0 and a component s_{Dairy} switched on by a smoothly varying dairying-onset field $D(i, t)$ — built by inverse-distance interpolation from six regional anchor dates (Mediterranean ~8200 BP through Baltic ~5600 BP) rather than four flat plateaus, so the covariate has genuine spatial texture. Per-region multipliers m let the north run hotter than the south, and the migration term exchanges frequency with adjacent land cells. Here is the model's actual geography — the land cells it evolves on, coloured by when dairying arrives in each:

```

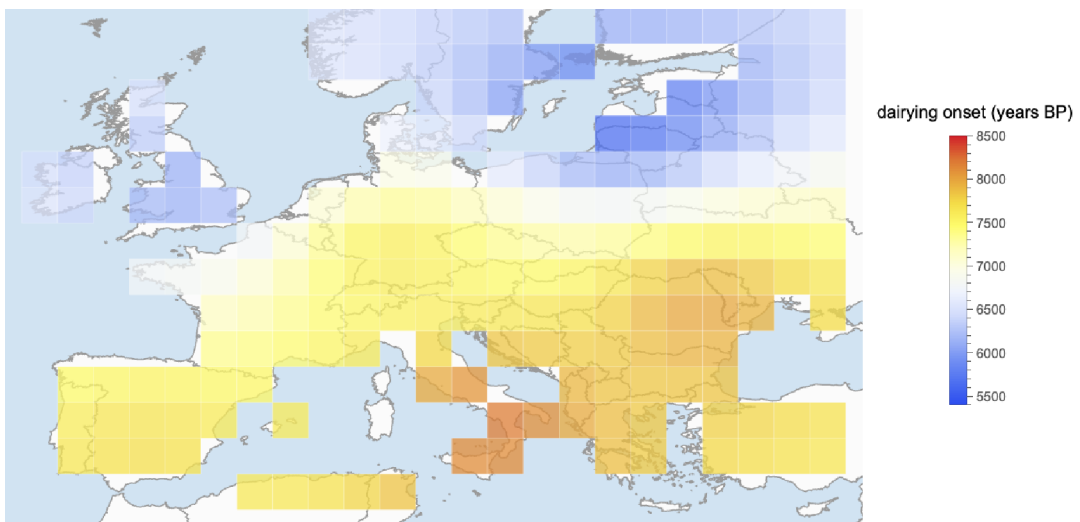
In[ ]:= grid = BuildEuropeGrid[];
Length[grid]

Out[ ]:=
191

In[ ]:= Legended[
  GeoGraphics[
    Flatten@Table[
      {GeoStyling[Opacity[0.75,
        ColorData["TemperatureMap"][Rescale[c["DairyingOnsetBP"], {5400, 8500}]]],
      EdgeForm[Directive[White, Opacity[0.6], AbsoluteThickness[0.3]]],
      GeoPolygon[{{c["Latitude"] - 1, c["Longitude"] - 1}, {c["Latitude"] - 1, c["Longitude"] + 1},
        {c["Latitude"] + 1, c["Longitude"] + 1}, {c["Latitude"] + 1, c["Longitude"] - 1}}],
      {c, grid}],
    GeoRange -> {{34, 63}, {-12, 36}}, GeoProjection -> "Equirectangular",
    GeoBackground -> GeoStyling[{"CountryBorders",
      "Land" -> GrayLevel[0.985], "Ocean" -> RGBColor[0.82, 0.89, 0.95]}],
    ImageSize -> 640],
    BarLegend[{{ColorData["TemperatureMap"][Rescale[#, {5400, 8500}]] &}, {5400, 8500}},
      LegendLabel -> "dairying onset (years BP)"]

Out[ ]:=

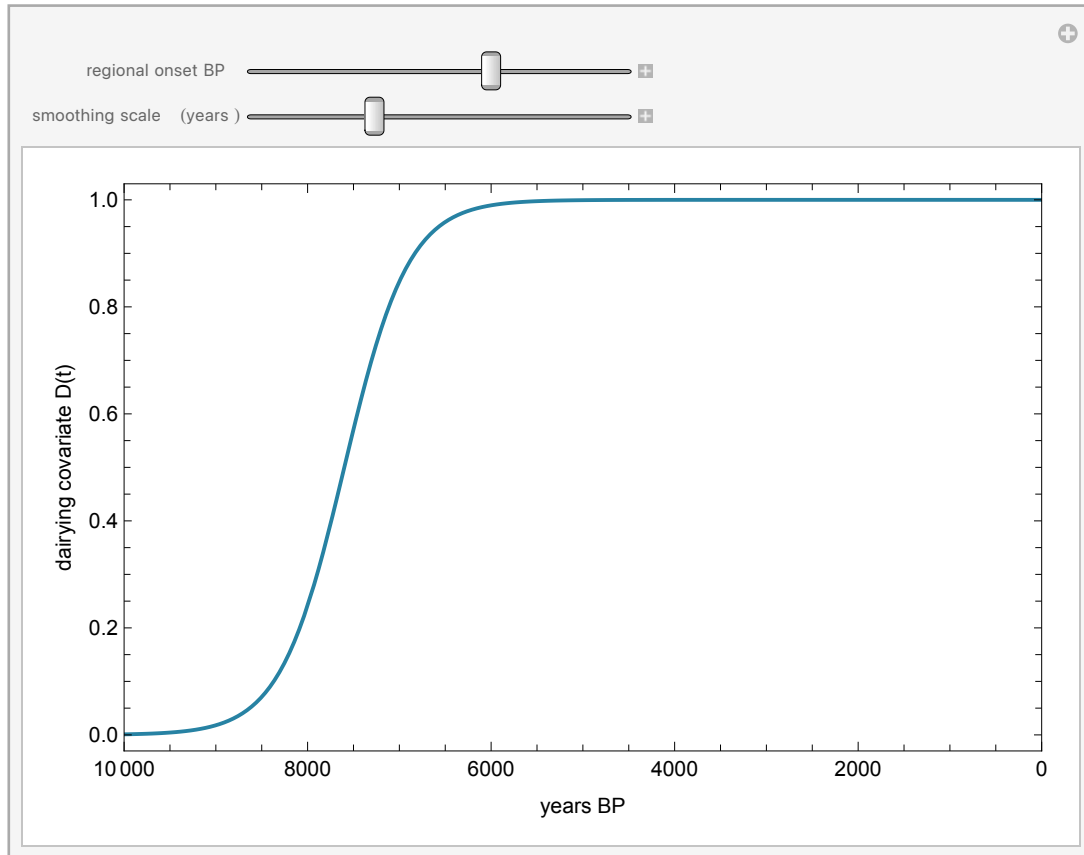
```



The model's spatial skeleton: the 2° land cells (sea cells are removed by an elevation test, so diffusion respects coastlines — Britain stays connected across the Dover cells, Ireland is an island in the adjacency too), coloured by the smooth dairying-onset field. Red is early dairying (Mediterranean and the southeast, ~8,200 BP), blue is late (the Baltic, ~5,600 BP). This field is what the sDairy component of selection switches on, cell by cell, as the simulation runs.

The covariate's shape in time is worth ten seconds of interaction too:

```
In[*]:= DairyCovariateExplorer[]
Out[*]=
```



One forward simulation of this model costs about 70 ms:

```
In[*]:= AbsoluteTiming[
  SimulateSpatialTrajectory[
    <|"InitialFrequency" -> 0.003, "SelectionBase" -> 0.002,
      "SelectionDairying" -> 0.02, "Migration" -> 0.003|>, grid];
][[1]]
Out[*]=
0.059114
```

— and that number is the entire strategy. When a simulator is this cheap, you do not need to linearize, approximate, or hand-wave the likelihood: you can afford tens of thousands of honest forward runs and let simulation-based inference do the rest. The model deliberately keeps demography implicit; it is a reaction–diffusion caricature whose job is to let the *data* say how much spatial structure, movement, and dairying-modulation they can constrain — not to smuggle conclusions in through the architecture.

External asset. The Neolithic dairying scene was generated outside Wolfram Language with OpenAI's gpt-image-2 model; the prompts and the one-off Python driver are checked into `docs/images/figures-generated/` of the project repository. It is a static illustration; there is no WL code to run for this figure.



5. Inference: sequential Monte Carlo ABC

Approximate Bayesian computation asks the only question a simulator can answer: which parameters produce synthetic data that look like the real data? "Look like" is made precise through summary statistics; here, two families. First, the called-allele-weighted regional time-binned frequencies (37 bins). Second — because a spatial model scored only on regional aggregates can never learn about migration — each sample is matched to its nearest grid cell and time step, and pooled north-south and west-east frequency contrasts over the last 4,000 years are computed *identically* for observed alleles and model expectations. The distance is a weighted root-mean-square over all of it.

Plain rejection ABC wastes almost every simulation once the tolerance gets interesting, so the fit uses sequential Monte Carlo: a population of 400 particles is filtered through 5 generations of shrinking tolerance (each generation's ϵ is the median of the previous population's distances), with Gaussian perturbation kernels and importance weights. One implementation lesson worth passing on: with a 10-dimensional box prior, perturbing in the raw coordinates throws ~97% of proposals out of the box (0.7 to the 10th ≈ 0.03). The sampler therefore works in logit-transformed coordinates, where a uniform prior becomes an exact product of standard logistic densities, every proposal is automatically valid, and the importance weights stay closed-form.

```

In[ ]:= smc = LoadOrRunSMCABC[Directory[], samples, grid];
Grid[
Prepend[
Table[{k, NumberForm[smc["EpsilonHistory"][[k]], {5, 4}},
NumberForm[100. smc["AcceptanceHistory"][[k]], {4, 1}},
NumberForm[smc["ESSHistory"][[k]], {5, 1}]], {k, Length[smc["EpsilonHistory"]]}],
Style[#, Bold] & /@ {"generation", "ϵ", "acceptance %", "ESS"}],
Frame -> All, FrameStyle -> GrayLevel[0.8], Spacings -> {1.5, 0.6},
Background -> {None, {GrayLevel[0.93], None}}]

```

Out[]:=

generation	ϵ	acceptance %	ESS
1	0.1549	100.0	400.0
2	0.0667	25.0	20.2
3	0.0599	25.0	18.9
4	0.0513	16.7	22.0
5	0.0443	12.5	16.0

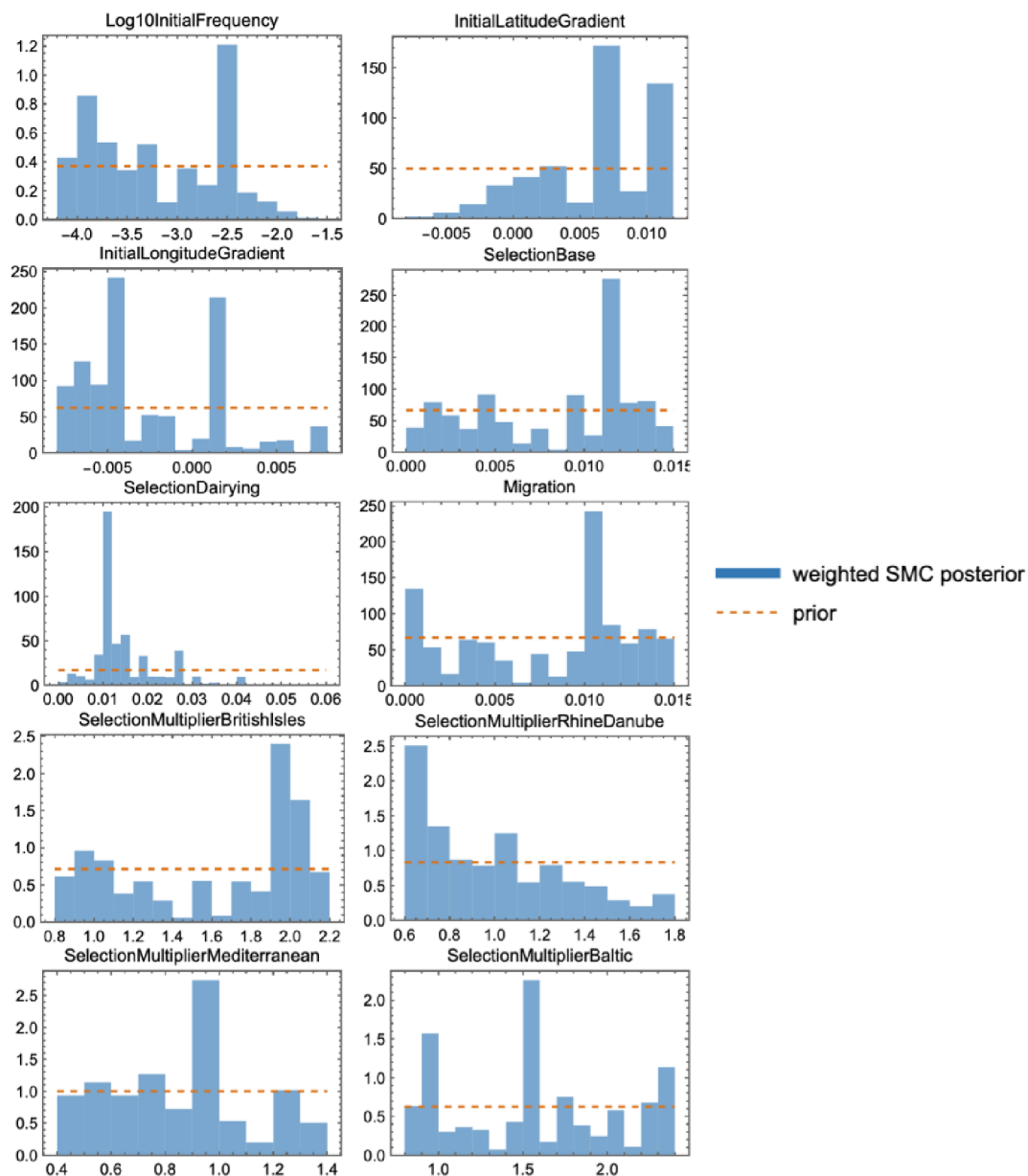
The tolerance falls from 0.155 (the prior median — i.e. no information) to 0.044 over 10,000 simulations. The final effective sample size, 16.0 of 400, deserves a sentence rather than a footnote: importance-weight degeneracy is the known cost of this weight formula whenever the posterior is much tighter than the prior, every interval in this post is a weighted quantile and correspondingly approximate, and the honest cure (an SMC sampler with MCMC rejuvenation moves) is on the to-do list in §11. Reporting ESS is what separates a posterior from a scatter plot.

```

In[ ]:= draws = ResamplePosterior[smc, 100];
smcOutputs = ExportSMCOutputs[Directory[], samples, grid, smc, draws];
Import[smcOutputs["ParameterFigure"]]

Out[ ]:=

```



Weighted SMC posterior (blue) against the flat prior (dashed orange) for all ten parameters. Prior overlays are the fastest honesty check in Bayesian workflow: one glance separates the dimensions the data actually moved (initial frequency, dairying-modulated selection, the northern multipliers) from the ones still wearing their priors (the initial spatial gradients).

6. What the posterior says

```
In[ ]:= quantiles = PosteriorParameterQuantiles[smc];
Grid[
Prepend[
Table[With[{q = SelectFirst[quantiles, #["Parameter"] === p &]},
{p, NumberForm[q["Median"], {6, 4}],
Row[{{"", NumberForm[q["Lower95"], {6, 4}], "", NumberForm[q["Upper95"], {6, 4}], ""}],
{p, {"Log10InitialFrequency", "SelectionBase", "SelectionDairying", "Migration",
"SelectionMultiplierBritishIsles", "SelectionMultiplierRhineDanube",
"SelectionMultiplierMediterranean", "SelectionMultiplierBaltic"}}],
Style[#, Bold] & /@ {"parameter", "median", "95% interval"}],
Frame -> All, FrameStyle -> GrayLevel[0.8], Spacings -> {1.5, 0.6},
Background -> {None, {GrayLevel[0.93], None}}]
```

Out[]:=

parameter	median	95% interval
Log10InitialFrequency	-3.2777	[-4.0983, -2.1111]
SelectionBase	0.0104	[0.0005, 0.0147]
SelectionDairying	0.0116	[0.0024, 0.0359]
Migration	0.0104	[0.0003, 0.0145]
SelectionMultiplierBritishIsles	1.8433	[0.8586, 2.1679]
SelectionMultiplierRhineDanube	0.9335	[0.6538, 1.7165]
SelectionMultiplierMediterranean	0.8968	[0.4377, 1.3820]
SelectionMultiplierBaltic	1.5672	[0.8188, 2.3750]

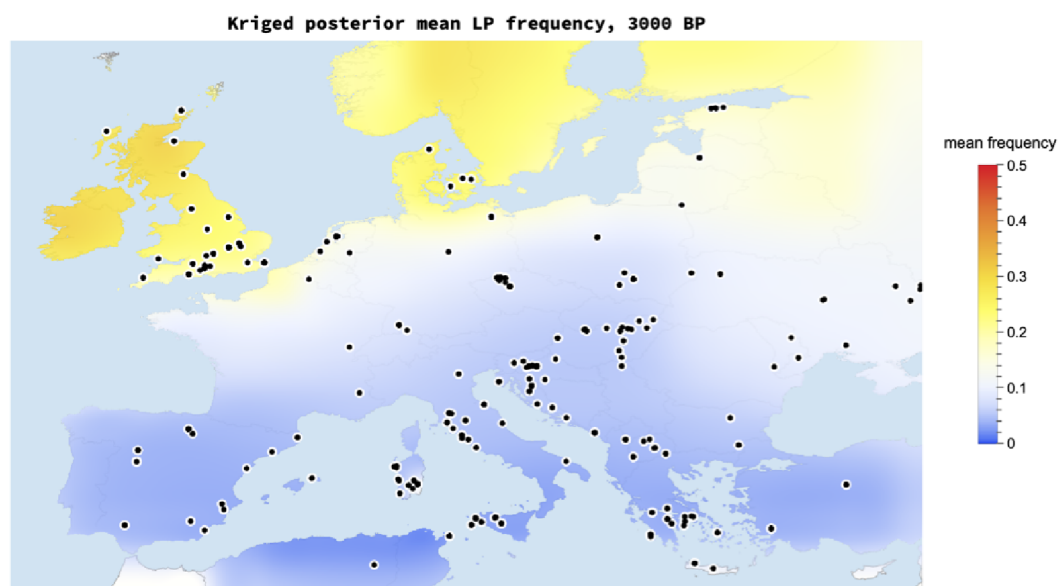
Three readings, in decreasing order of confidence. **First**, the total effective selection in a northern dairying cell at the posterior median — $s_0 + s_{\text{Dairy}} \times \text{multiplier} \approx 0.032$ per generation — sits squarely in the published few-percent range, with the northern multipliers pulled above one exactly where the logistic slopes were steepest. **Second**, the posterior does *not* cleanly separate dairying-modulated from baseline selection. This is not a failure of the sampler; it is the geometry of the problem. The dairying covariate saturates to one within a few centuries of onset, so for most of the simulated period s_{Dairy} acts almost exactly like uniform selection; only the early Neolithic carries contrast between the two. That the ancient genotypes accommodate both a mostly-uniform and a mostly-dairy-modulated reading is this little model's echo of Evershed et al.'s conclusion at Nature scale: milk exploitation is not required to explain the trajectory. **Third**, migration is only weakly identified — §9 quantifies exactly how weakly, instead of letting a prior masquerade as a discovery.

7. The posterior on the map – and its uncertainty

Every map below is a posterior functional: 100 equally-weighted posterior draws, each pushed through the forward model, summarized per cell, then interpolated for display by ordinary kriging (range 3.5°, nugget 0.02) and cut to the coastline. Kriging here is a display courtesy, not an inferential claim — the 2° cells remain the units that saw data. Colors are the TemperatureMap scale with a square-root stretch so the long rare-allele centuries stay legible.

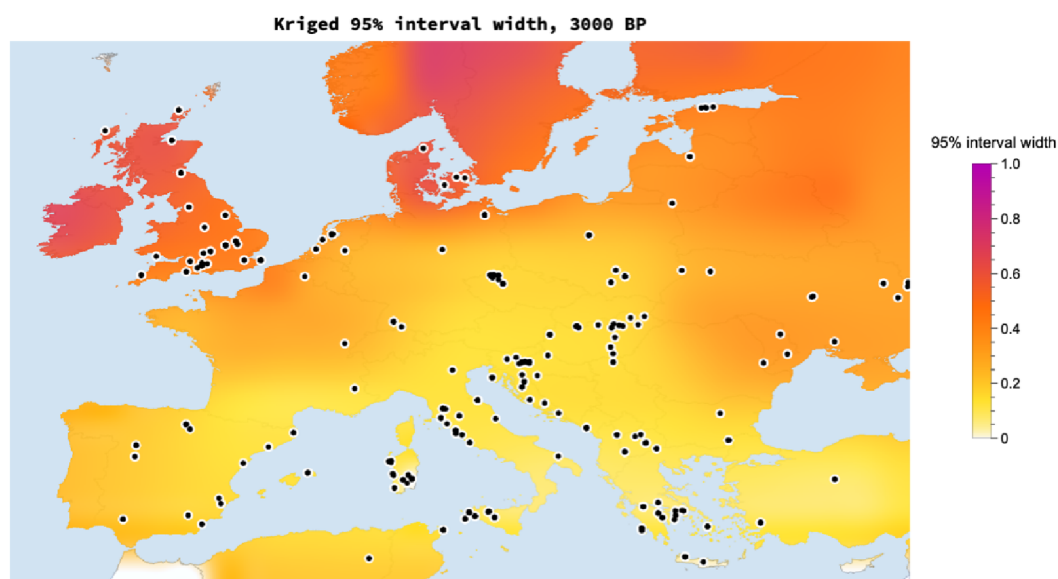
```
In[ ]:= spatialOutputs = ExportSpatialVisualizations[Directory[], samples, grid, draws];
Import[spatialOutputs["MeanMap"]]
```

```
Out[ ]:=
```



```
In[ ]:= Import[spatialOutputs["UncertaintyMap"]]
```

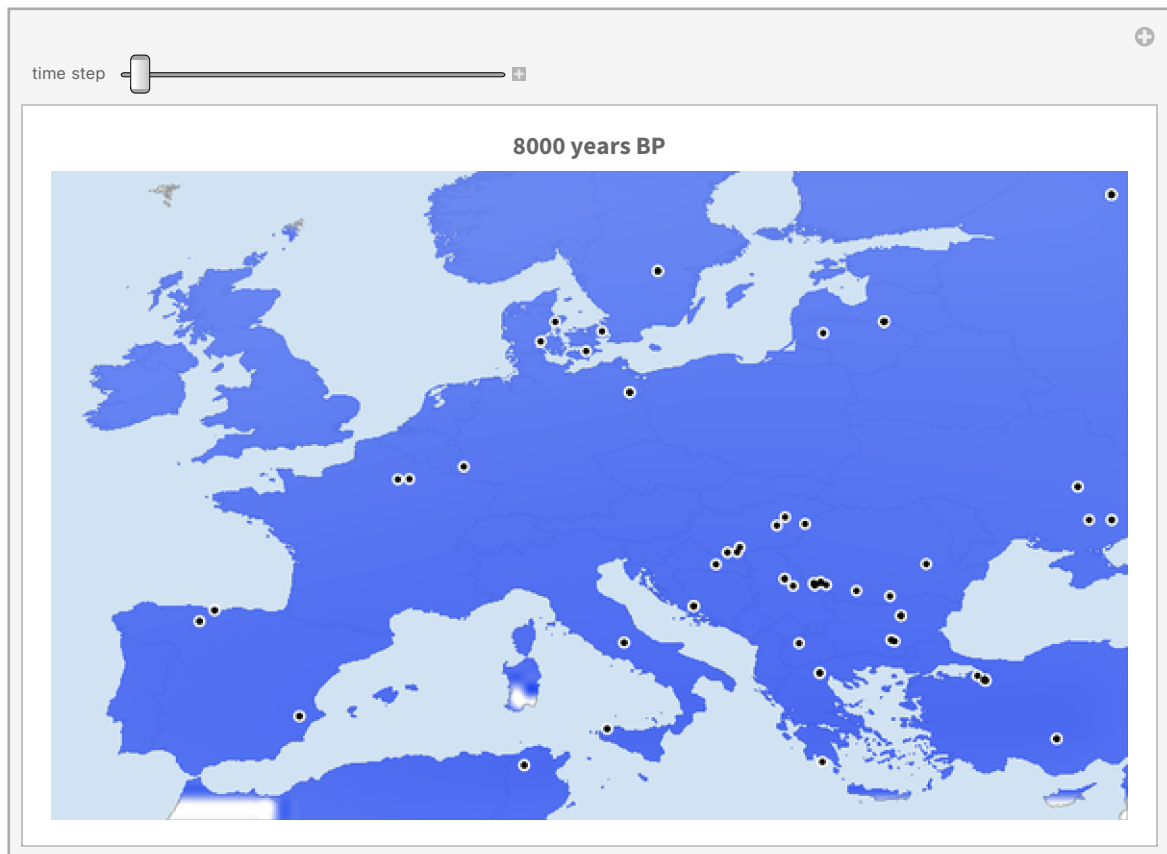
```
Out[ ]:=
```



Posterior mean (top) and 95% credible-interval width (bottom) at 3000 BP. The two maps are a pair on purpose: the second is the error bar of the first, rendered at equal visual rank. The interval is widest exactly where the sampling map of §1 is emptiest — the model has the good manners to be unsure where it has seen nothing.

And because a static pair understates how the story unfolds, step through it yourself — each frame is the posterior mean with the ancient samples of that moment:


```
In[*]:= SpatialTimeExplorer[samples, grid, draws]
Out[*]=
```



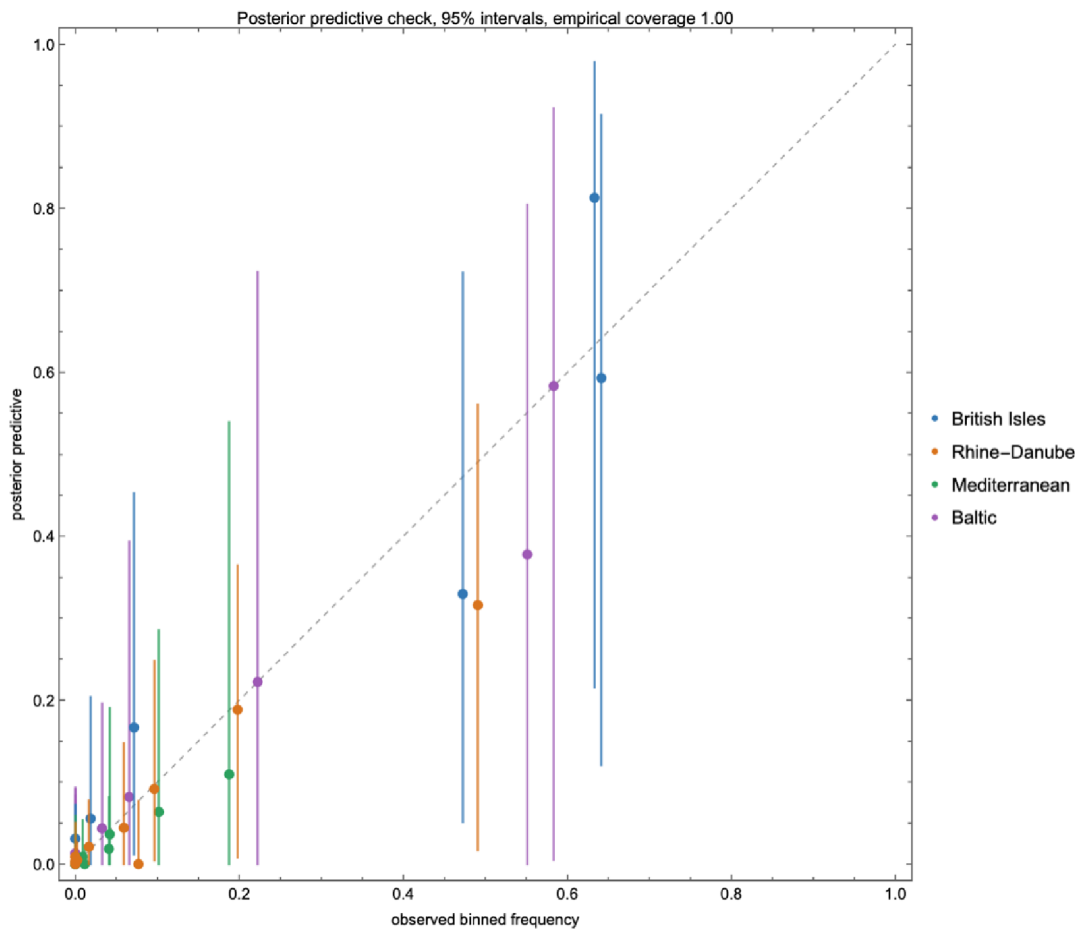
8. Does it actually predict anything?

A posterior that merely redescribes its training data is a compression algorithm, not a model. Three checks, in increasing order of severity.

Posterior predictive. Push posterior draws through the model, sample binomial allele counts at each bin's actual sample size, and ask how often the observed frequency lands inside the 95% band: 100% of the 37 bins. Bins with two called alleles correctly get bands covering nearly everything — wide honesty beats narrow fiction.

```
In[*]:= Import[smcOutputs["PosteriorPredictiveFigure"]]
```

```
Out[*]:=
```



Held-out regions. Refit the entire SMC from scratch four times, each time deleting one region's samples, then predict the deleted region:

```
In[*]:= cv = LoadOrRunCrossValidation[Directory[], samples, grid];
Dataset[cv]
```

```
Out[*]:=
```

HeldOutRegion	HeldOutBins	RMSE	Coverage95	FinalEpsilon	TotalSimu
British Isles	7	0.226313	0.857143	0.0416138	2400
Rhine-Danube	10	0.0555733	1.0	0.0616102	2100
Mediterranean	10	0.0618259	1.0	0.053258	2400
Baltic	10	0.0634689	1.0	0.0505987	2100

Held-out time — the killer test. Train only on samples at or older than the 3,000 BP bin edge (4459 of them) and predict the 12 most recent bins. The intervals still cover, but the posterior median under-predicts the final rise badly (RMSE 0.33). Far from being embarrassing, this is the pipeline independently rediscovering Burger et al.'s headline: the evidence for *strong, late* selection is concentrated in the last three millennia. Delete those millennia and the model, sensibly, refuses to believe in them.

```
In[ ]:= timeSlice = LoadOrRunTimeSliceValidation[Directory[], samples, grid];
Dataset[{timeSlice}]
```

```
Out[ ]:=
```

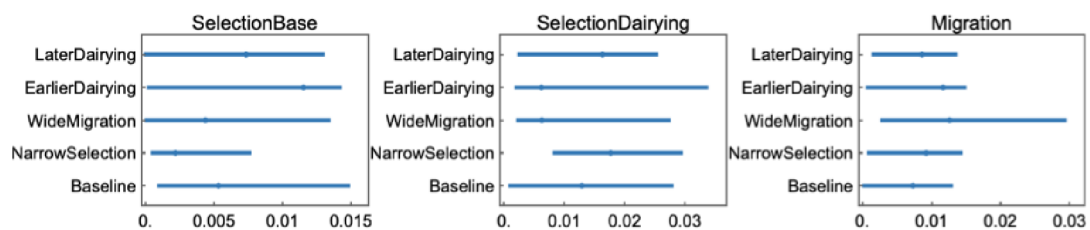
CutBP	HeldOutBins	TrainingSamples	RMSE	Coverage95	TotalSimulations
3000	12	4459	0.325715	0.916667	2100

9. Which conclusions survive the priors?

Five complete refits: baseline; halved selection priors; doubled migration prior; and every dairying onset shifted 400 years earlier or later. The dairying-modulated selection component stays positive in all five (medians 0.006–0.018) — though \$11 will temper how much of that is data rather than the non-negative prior. The migration posterior tracks whichever prior it is given (median 0.007 baseline → 0.013 under the widened prior): with 191 cells and regionally-pooled summaries, movement is weakly identified, and the honest statement is exactly that.

```
In[ ]:= sens = LoadOrRunSensitivity[Directory[], samples];
Import[FileNameJoin[{Directory[], "figures", "generated", "sensitivity_intervals.png"}]]
```

```
Out[ ]:=
```



10. Where did the allele come from? Origin versus selection

Everything so far answers "where did selection push hardest?" — which is a different question from "where did the allele come from?", and the two are easy to conflate when watching the animation. The classic origin estimate is **Itan et al. (2009)**, whose demic coevolution simulations placed the start of the lactase-persistence/dairying story around 7,500 years ago in the zone between the central Balkans and central Europe — roughly the Linearbandkeramik world of modern Hungary, Slovakia, and Austria. What do the ancient genotypes in this dataset say about first appearances?

```

In[ ]:= earliest = TakeLargestBy[
  Select[samples, #["DerivedAlleles"] > 0 && #["MeanDateBP"] <= 10000 &],
  #["MeanDateBP"] &, 10];
Grid[
  Prepend[
    {Round[#["MeanDateBP"]], #["Country"], #["Latitude"], #["Longitude"],
      #["RS4988235Genotype"], #["Publication"]} & /@ earliest,
    Style[#, Bold] & /@ {"years BP", "country", "lat", "lon", "call", "publication"}],
  Frame -> All, FrameStyle -> GrayLevel[0.8], Spacings -> {1.2, 0.6},
  Background -> {None, {GrayLevel[0.93], None}}]

```

Out[]:=

years BP	country	lat	lon	call	publication
8813	Netherlands	52.01	3.53	A	PosthKrauseNature2023
8752	USA	46.2114	-119.136	AA	RasmussenWillerslevNature2015
8500	Turkey	37.6661	32.8257	AA	YüncüSomelScience2025
7550	China	41.2942	114.23	A	ZhangCuiSciBull2025
7125	Ukraine	48.1213	35.0902	A	NikitinReichNature2025
6785	Russia	42.999	133.095	AA	WangPosthCurrBiol2023
6569	Bulgaria	43.2131	27.8644	A	MathiesonReichNature2018
6402	France	47.97	3.56	A	RivollatHaakNature2023
4850	Russia	45.5181	44.619	A	LazaridisReichNature2025
4614	Russia	47.22	40.1	A	LazaridisReichNature2025

```

In[ ]:= GeoGraphics[
  Flatten[Table[{RGBColor[0.8, 0.15, 0.1], PointSize[0.013],
    Point[GeoPosition[{e["Latitude"], e["Longitude"]}]],
    Black, Text[Style[ToString[Round[e["MeanDateBP"]]] <> " BP", 9.5, Bold],
      GeoPosition[{e["Latitude"] + 1.1, e["Longitude"]}]]],
    {e, earliest}], 1],
  GeoRange -> {{34, 63}, {-14, 72}}, GeoProjection -> "Equirectangular",
  GeoBackground -> GeoStyling[{"CountryBorders",
    "Land" -> GrayLevel[0.97], "Ocean" -> RGBColor[0.85, 0.9, 0.95]}],
  ImageSize -> 640]

```

Out[]:=



The oldest derived-allele calls within the modelled era ($\leq 10,000$ BP), from the AADR v66 release. Read this table with suspicion, not reverence: most rows are single sequencing reads, and G→A is the very transition that post-mortem damage produces, so an isolated early “carrier” is as likely chemistry as biology — which is why the model fits an explicit error rate

rather than taking the table at face value. The cautionary tale is what happened to the previous version of this table: under the 2021-era data behind our first fit, the three oldest carriers formed a tidy southeastern arc (Bulgaria ~6,560 BP, Ukraine ~5,600 BP, Romania ~5,390 BP). Current data dissolve two of the three — the Ukrainian individual has been redated to ~3,985 BP and the Romanian heterozygote is now called GG from the diploid genotypes — leaving one depth-2 heterozygote from Bulgaria carrying what was once a whole origin narrative.

The hero animation cannot show any of this, *by construction*: its model starts at 10,000 BP with a near-uniform initial frequency, has no mutation event, and — with migration weakly identified — moves essentially by in-place logistic growth. Britain lights up first there because its late samples demand the fastest growth, not because the allele originated there. So let us stop hand-waving and put the origin *into* the model: replace the uniform start with a point source. Six new parameters: origin latitude, longitude and time (time prior 5,000–10,000 BP, wide enough that both Itan et al.'s 7,441 BP and the ~6,000 BP rise date of Irving-Pease et al. live inside it); the injection frequency; a genotype-error rate $\epsilon \in [0, 0.02]$, because G→A is exactly the transition that post-mortem deamination fakes, and a model without an error channel is forced to treat every damaged Palaeolithic read as a real carrier; and a dairying lead time. That last one is the scientific commitment, made explicit and *fitted* rather than hidden in the prior: the source may not precede the local dairying onset by more than the lead (prior 0–2,000 years; origins after local onset are unrestricted — the constraint is deliberately asymmetric), because the thing being located is the start of the *selection-driven rise*, not the first mutation event — the same gene–culture coupling Itan et al. build into their model. Without it, a drift-free simulator lets a vanishingly small injection idle for millennia in pre-dairying forager regions. Migration gets a wider prior than the main model (a travelling wave has to actually travel: $m \in [0.02, 0.3]$ per generation, capped where the mixing map saturates), the integrator uses exact per-step logistic growth with exponential mixing, and diagonal adjacency lets the wave cross the Dover strait. The same SMC machinery then turns "where did it start?" into a posterior distribution:

```

In[ ]:= originSmc = LoadOrRunOriginSMCABC[Directory[], samples, grid];
Grid[
Prepend[
Table[With[{q = SelectFirst[PosteriorParameterQuantiles[originSmc], #["Parameter"] === p &]},
{p, NumberForm[q["Median"], {6, 3}],
Row[{{"["", NumberForm[q["Lower95"], {6, 3}], ", ", NumberForm[q["Upper95"], {6, 3}], "]"}}],
{p, {"OriginLatitude", "OriginLongitude", "OriginTimeBP",
"Log10InjectFrequency", "Migration", "SelectionDairying",
"DairyingLeadYears", "CallErrorRate"}}],
Style[#, Bold] & /@ {"parameter", "median", "95% interval"}],
Frame -> All, FrameStyle -> GrayLevel[0.8], Spacings -> {1.5, 0.6},
Background -> {None, {GrayLevel[0.93], None}}]

```

Out[]:=

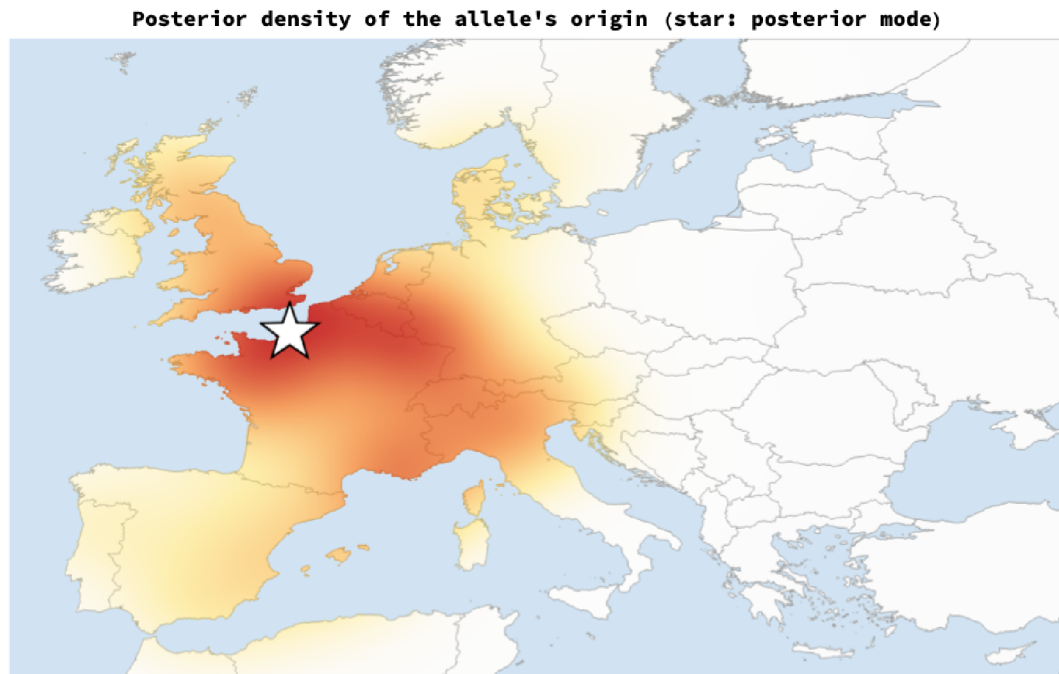
parameter	median	95% interval
OriginLatitude	48.320	[37.480, 57.895]
OriginLongitude	2.288	[-8.731, 11.313]
OriginTimeBP	6650.340	[5120.860, 8844.580]
Log10InjectFrequency	-1.964	[-2.985, -1.073]
Migration	0.164	[0.032, 0.292]
SelectionDairying	0.043	[0.023, 0.058]
DairyingLeadYears	1459.830	[144.328, 1958.450]
CallErrorRate	0.012	[0.000, 0.019]

```

In[ ]:= OriginDensityMap[originSmc]

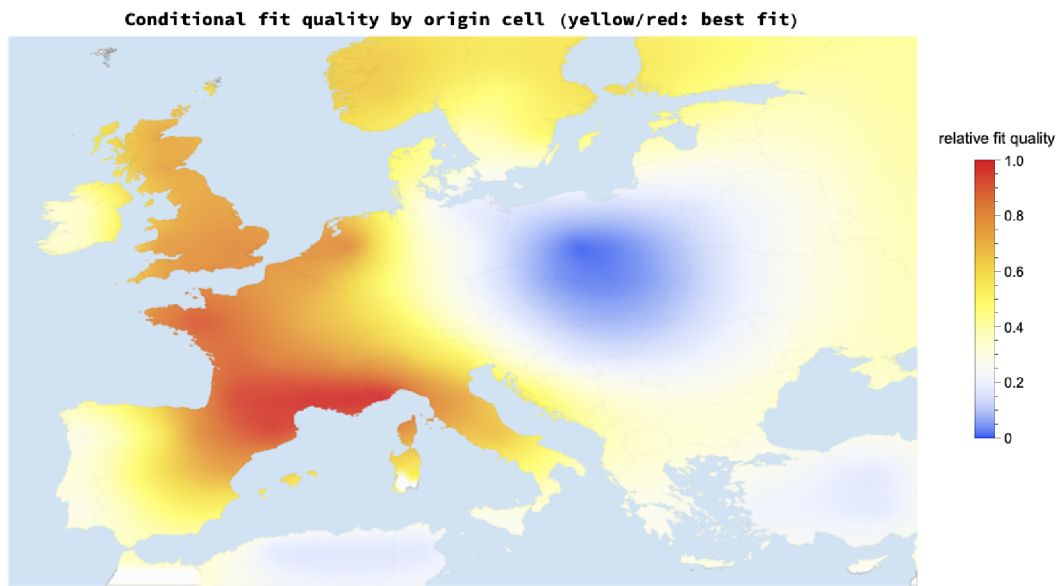
```

Out[]:=



The reverse-diffusion result, in the spirit of Itan et al.'s famous origin map: weighted posterior density of the point source's location, land-masked. Read the spread, not the star. §11 shows that this posterior is broad and multimodal enough that its mode is not a stable quantity — move the kernel bandwidth or reshuffle the resampling seed and the starred cell jumps between southwestern France, northern Germany and Iberia, a range of some 1,500 km. The star marks the modal cell of one particular kernel estimate; it is not a location estimate, and §11 makes the case that this posterior has no defensible point summary of location at all.

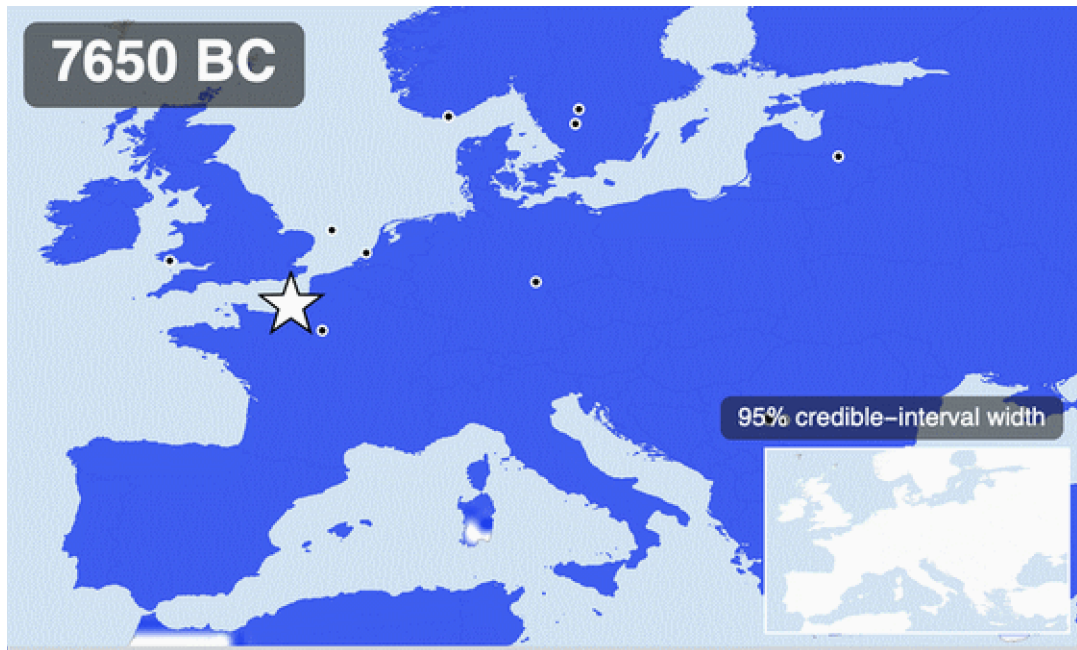

```
In[ ]:= OriginFitSurfaceMap[samples, grid, originSmc]
Out[ ]:=
```



A prior-free cross-check on the same question: every non-origin parameter is pinned at its posterior median, the point source is placed in each land cell in turn, and the map is coloured by how well the forward simulation then matches the ancient samples (warmer = better fit). Read it honestly, in both directions. The well-fitting region is broad: the ancient data constrain the origin's latitude band far more sharply than its longitude, and because the northwest of the map is sampled most densely, the conditional optimum leans that way. But what the scan rules out is just as informative — Iberia, the southern Balkans, and the far northeast fit distinctly worse — and the posterior concentration above sits comfortably inside the preferred band. Where the joint posterior differs from this scan — most visibly in pulling mass away from regions where dairying arrived late — the difference is the gene–culture coupling and the joint fitting of origin time and injection frequency, not the origin prior, which is uniform over the whole map.

And the forward story from that fitted origin — the travelling wave the hero animation could not show — as its own video (the star marks the posterior modal origin; the inset again tracks the 95% credible-interval width):

```
In[*]:= originSpread = ExportOriginSpread[Directory[], samples, grid, originSmc];
Out[*]=
```



Forward simulation from the fitted point source, rendered by the call above: posterior mean over 100 origin-model draws, 100-year interpolated steps. The H.264 version lands in figures/generated/origin_spread.mp4.

Read the origin posterior with the same discipline as everything else here. The location information comes from a handful of early single-heterozygote calls scattered over an unevenly sampled map, filtered through a coarse deterministic wave model — so the credible region is wide, and it should be. What the fit adds beyond the anecdotal table above is that all three ingredients are now estimated *jointly*: an origin location and time, a wave that must actually reach Britain and the Baltic in time for their observed rises, and selection on top. Whether the posterior mass sits over the Carpathian Basin, the lower Danube, or the Pontic steppe is exactly the kind of statement this machinery exists to make — and unlike a caption, it comes with its uncertainty attached.

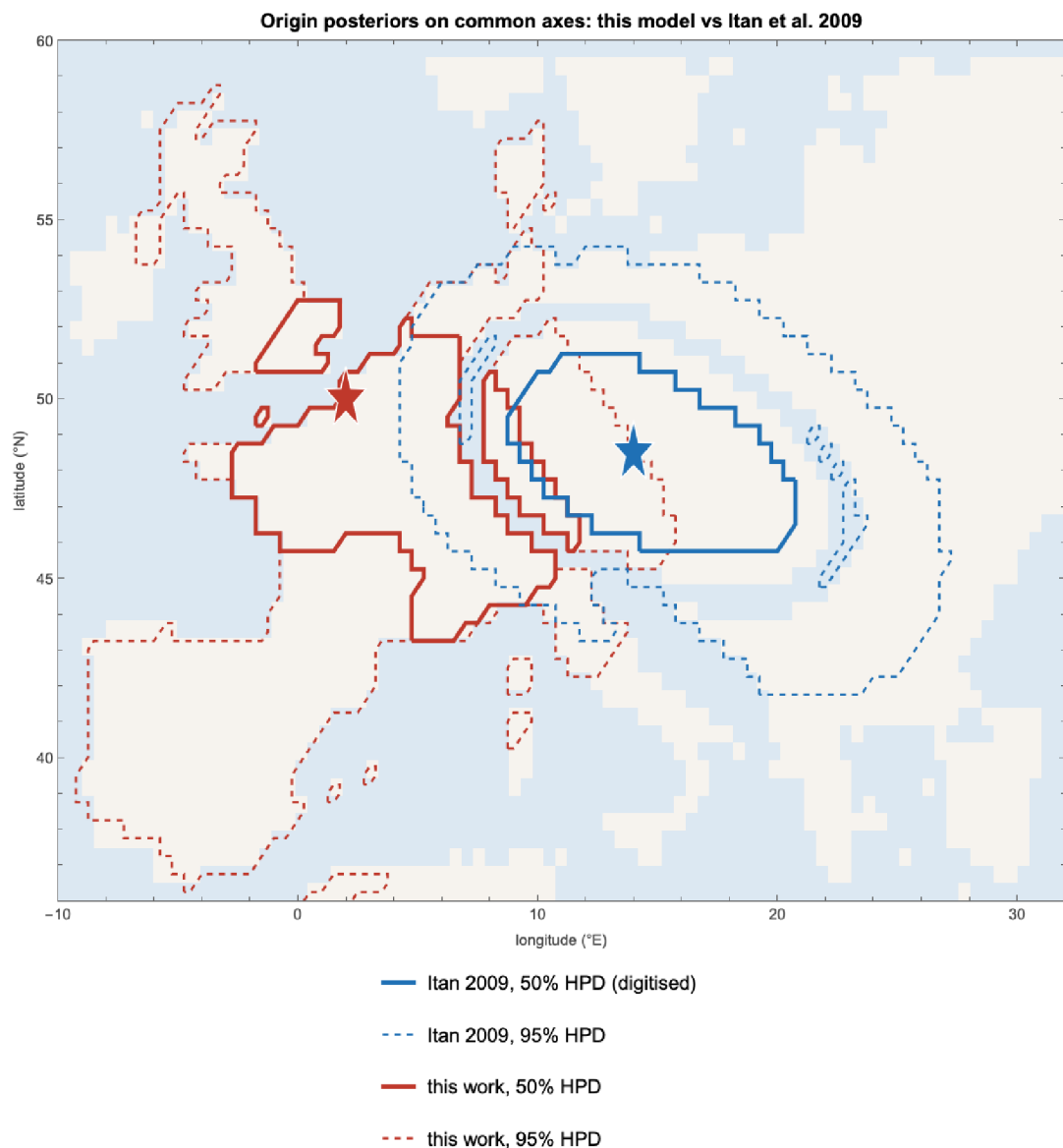
11. Checked against the published record

A model that only agrees with itself is not worth posting, so this section tests the fit against estimates it never saw. The benchmark is **Itan et al. (2009)** — a demic coevolution simulation driven by archaeological settlement dates and *modern* allele frequencies, with no ancient genotypes at all. Our fit uses *only* ancient genotypes. The two are largely — not entirely — independent: our gene-culture coupling consults the same archaeological dairying chronology that drives their simulation, so the origin-date comparison shares an input, while the genetic data are disjoint. That still makes the comparison worth doing — and worth doing *numerically*, not by putting two pictures next to each other and inviting the eye to find agreement. Two maps that share a warm patch can still be quantitatively far apart, and here they are.

So the first step is to turn their figure back into numbers — in Wolfram. Their Fig 3 carries tick marks at known latitudes and longitudes, so the pixel grid can be georeferenced exactly (the parallels are horizontal; the meridians converge northward at 35.85 px/degree along the top frame and 42.15 along the bottom), and the filled contour bands run along a blue-cyan-green-yellow-red ramp

that inverts analytically. `ImportItanFig3Density` in the package does all of this in about forty lines of image processing; an independent Python implementation (`scripts/digitise_itan_fig3.py`) agrees with it to a mean absolute difference of 0.002 on the unit ramp scale. Two caveats are stated rather than hidden. First, the paper publishes no numeric colour scale, so equal density spacing between contour bands is an assumption; mode locations are robust to any monotone recolouring, but the overlap statistics below are approximate. Second, the paper's text localises the origin only qualitatively ("between the central Balkans and central Europe"); the coordinate 48.5°N, 14°E quoted in this section is the mode of our own georeferenced digitisation, not a number from their text. With both posteriors as numbers on one grid, the comparison becomes arithmetic:

```
In[*]:= OriginItanHPDComparisonMap[originSmc, Directory[]]
Out[*]=
```



Highest-posterior-density contours for both analyses on common axes — derived numeric contours only, no part of the published image reproduced. Itan et al.'s 50% region (blue, solid) is a compact ellipse over Bavaria, Bohemia and the Carpathian Basin. Ours (red, solid) sprawls from the Atlantic coast of France to Ukraine in several disconnected lobes, and our 95% region covers essentially the whole domain. This is the comparison the side-by-side view flattered.

```

In[ ]:= itanGrid = Import[FileNameJoin[{Directory[], "data", "processed", "itan2009_origin_density_digitised.c
itanRows = Normal[itanGrid];
lons = Union[#[[1]] & /@ itanRows]; lats = Union[#[[2]] & /@ itanRows];
itanAssoc = Association[{#[[1]], #[[2]]} -> #[[3]] & /@ itanRows];
originDraws2 = ResamplePosterior[originSmc, 2000];
kdeAt[lo_, la_] := Mean[Map[Exp[-0.5 (((lo - #[[1]] - "OriginLongitude")/2.6)^2 + ((la - #[[1]] - "OriginLatitude")/2.
cells = Keys[itanAssoc];
wArea = Cos[#[[2]] Degree] & /@ cells;
pl = (Values[itanAssoc] wArea); pl = pl/Total[pl];
pO = (kdeAt @@@ cells) wArea; pO = pO/Total[pO];
hpdMask[p_, lev_] := Module[{ord = Ordering[-p], c = 0., keep = {}},
  Do[c += p[[i]]; AppendTo[keep, i]; If[c >= lev, Break[], {i, ord}]; keep];
bc = Total[Sqrt[pl pO]];
Grid[{Style[#, Bold] & /@ {"statistic", "value", "reading"},
  {"Bhattacharyya coefficient", NumberForm[bc, {4, 3}], "1 = identical, 0 = disjoint"},
  {"our mass in Itan 50% region", NumberForm[Total[pO[[hpdMask[pl, 0.5]]]], {4, 3}], "vs 0.50 if we agreed"},
  {"our mass in Itan 95% region", NumberForm[Total[pO[[hpdMask[pl, 0.95]]]], {4, 3}], "vs 0.95 if we agree"},
  {"Itan mass in our 95% region", NumberForm[Total[pl[[hpdMask[pO, 0.95]]]], {4, 3}], "ours is broad enou
Frame -> All, FrameStyle -> GrayLevel[0.8], Spacings -> {1.4, 0.6},
Background -> {None, {GrayLevel[0.93], None}}]

```

Out[] =

statistic	value	reading
Bhattacharyya coefficient	0.421	1 = identical, 0 = disjoint
our mass in Itan 50% region	0.082	vs 0.50 if we agreed
our mass in Itan 95% region	0.306	vs 0.95 if we agreed
Itan mass in our 95% region	0.399	ours is broad enough to contain theirs

The honest verdict on location. If the two posteriors were describing the same belief, the mass we place inside their 50% and 95% regions would be about 0.50 and 0.95. It is nothing like that. The last row is the one that is easy to misread: their whole distribution sits inside our 95% region, but that is a statement about how vague we are, not how right. A posterior wide enough to contain every rival hypothesis has not corroborated any of them.

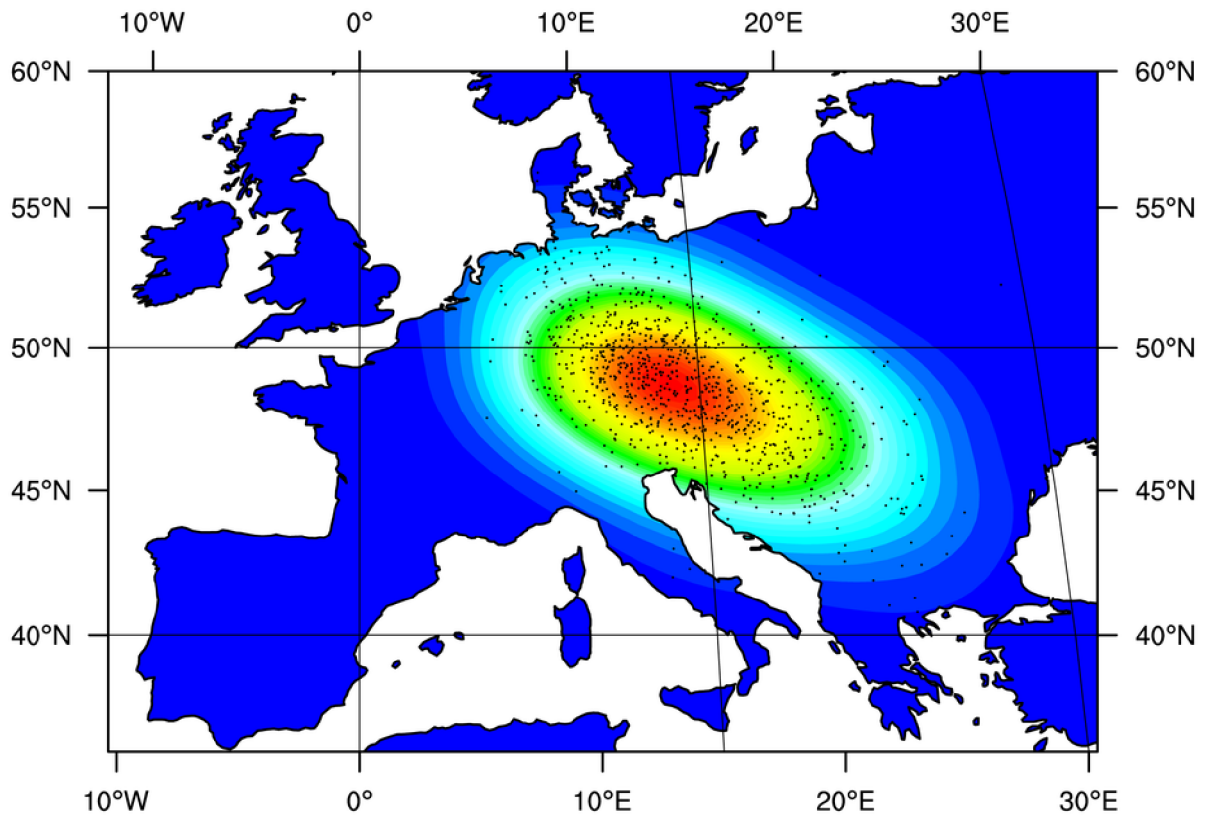


Figure 3 of: Itan Y, Powell A, Beaumont MA, Burger J, Thomas MG (2009), “The Origins of Lactase Persistence in Europe”, *PLoS Computational Biology* 5(8): e1000491, doi:10.1371/journal.pcbi.1000491. © 2009 Itan et al., distributed under the Creative Commons Attribution (CC BY) licence (creativecommons.org/licenses/by/4.0), which permits unrestricted reproduction provided the original author and source are credited. Reproduced here unmodified; separately, the figure was digitised into the numeric density field used above (a derived work under the same licence). It shows the posterior density of the origin of the lactase-persistence/dairying coevolution in their demic model, with the high-density region running from Bavaria/Bohemia toward the Carpathian Basin — the “Hungary result” of the popular retellings.

```

In[ ] := originDraws = ResamplePosterior[originSmc, 2000];
sTotal = Quantile[Map[##["SelectionBase"] + ##["SelectionDairying"] &, originDraws], {0.025, 0.5, 0.975}];
oRow[p_] := SelectFirst[PosteriorParameterQuantiles[originSmc], ##["Parameter"] === p &];
fq[p_, d_] := With[{q = oRow[p]}, {ToString[NumberForm[q["Median"], {8, d}]],
  "[" <> ToString[NumberForm[q["Lower95"], {8, d}]] <> ", " <> ToString[NumberForm[q["Upper95"], {8,
Grid[
{Style[#, Bold] & /@ {"quantity", "this model", "95% interval", "published", "source"},
Join[{"origin latitude (°N)", fq["OriginLatitude", 1], {"≈48.5 (mode)", "Itan 2009, Fig 3"}],
Join[{"origin longitude (°E)", fq["OriginLongitude", 1], {"≈13 (mode)", "Itan 2009, Fig 3"}],
Join[{"origin date (years BP)", fq["OriginTimeBP", 0], {"7441 [6256, 8683]", "Itan 2009"}],
Join[{"selection, dairying term", fq["SelectionDairying", 3], {"0.0953 [0.0518, 0.159]", "Itan 2009 (dairy
{"selection, total in dairying cells", ToString[NumberForm[sTotal[[2]], {8, 3}]],
  "[" <> ToString[NumberForm[sTotal[[1]], {8, 3}]] <> ", " <> ToString[NumberForm[sTotal[[3]], {8, 3}]] <>
  "0.014–0.15", "Bersaglieri 2004"},
{"", "", "", "≈0.06 (last 3 kyr)", "Burger 2020"}],
Frame -> All, FrameStyle -> GrayLevel[0.8], Spacings -> {1.2, 0.6},
Background -> {None, {GrayLevel[0.93], None}}]

```

Out[] =

quantity	this model	95% interval	published	source
origin latitude (°N)	48.3	[37.5, 57.9]	≈48.5 (mode)	Itan 2009, Fig 3
origin longitude (°E)	2.3	[−8.7, 11.3]	≈13 (mode)	Itan 2009, Fig 3
origin date (years BP)	6650.	[5121., 8845.]	7441 [6256, 8683]	Itan 2009
selection, dairying term	0.043	[0.023, 0.058]	0.0953 [0.0518, 0.159]	Itan 2009 (dairymen)
selection, total in dairying cells	0.052	[0.031, 0.067]	0.014–0.15	Bersaglieri 2004
			≈0.06 (last 3 kyr)	Burger 2020

Parameter by parameter. The date lands inside the published envelope: our median of 6650 BP [5121, 8845] against Itan et al.'s 7,441 [6,256–8,683] — overlapping intervals, with our posterior leaning later, toward the ~6,000 BP rise date that Irving-Pease et al. (2024) infer from 1,664 imputed ancient genomes. Selection is order-of-magnitude consistent rather than “inside every range”: the total in the table sits inside Bersaglieri's wide 0.014–0.15 and near Burger's ≈0.06, but below Itan's dairymen-only interval [0.0518, 0.159] and above Irving-Pease's genome-wide 0.0194 — and these four are not the same quantity (they differ in dominance, conditioning, and time-averaging), so exact containment was never the right test. Location is where honest trouble lives, and marginal medians hide it: a marginal median is a poor summary of a broad multimodal joint posterior, and quoting one here would manufacture agreement the joint distribution does not support.

One more check, at the other end of time: extrapolate the fitted uniform-start model to the present (0 BP, i.e. 1950) and compare with today's measured rs4988235-A frequencies from the 1000 Genomes Project at the four sampled populations that sit on our grid:

```

In[ ]:= modern = {
  {"Britain (GBR)", 52.5, -1.5, 0.720},
  {"Iberia (IBS)", 40.4, -3.7, 0.458},
  {"Tuscany (TSI)", 43.5, 11.0, 0.089},
  {"Finland (FIN)", 61.0, 25.0, 0.591}};
present = PosteriorCellStats[draws, grid, {0}][0];
nearCell[la_, lo_] := First[Ordering[
  Map[EuclideanDistance[{la, lo Cos[la Degree]],
    {#[["Latitude"]], #[["Longitude"] Cos[#[["Latitude"] Degree]]} &, grid], 1]];
Grid[
  Prepend[
    Map[Function[row, With[{i = nearCell[row[[2]], row[[3]]}],
      {row[[1]], ToString[NumberForm[present["Mean"][[i]], {4, 2}]],
      "[" <> ToString[NumberForm[present["Lower95"][[i]], {4, 2}]] <> ", " <>
      ToString[NumberForm[present["Upper95"][[i]], {4, 2}]] <> "]"",
      ToString[NumberForm[row[[4]], {4, 2}]]], modern],
    Style[#, Bold] & /@ {"population", "model at 0 BP", "95% interval", "1000G rs4988235-A"}],
  Frame -> All, FrameStyle -> GrayLevel[0.8], Spacings -> {1.4, 0.6},
  Background -> {None, {GrayLevel[0.93], None}}]

```

Out[]:=

population	model at 0 BP	95% interval	1000G rs4988235-A
Britain (GBR)	0.80	[0.30, 0.98]	0.72
Iberia (IBS)	0.16	[0.00, 0.69]	0.46
Tuscany (TSI)	0.25	[0.00, 0.64]	0.09
Finland (FIN)	0.64	[0.00, 0.98]	0.59

```

In[ ]:= mae = Mean[Table[With[{i = nearCell[row[[2]], row[[3]]}],
  Abs[present["Mean"][[i]] - row[[4]]], {row, modern}]];
maeNull = Mean[Abs[0.5 - #[[4]]] & /@ modern];
Grid[{Style[#, Bold] & /@ {"predictor", "mean absolute error"},
  {"this model, medians at 0 BP", NumberForm[mae, {4, 3}]},
  {"constant 0.5 for everyone", NumberForm[maeNull, {4, 3}]},
  Frame -> All, FrameStyle -> GrayLevel[0.8], Spacings -> {1.6, 0.6},
  Background -> {None, {GrayLevel[0.93], None}}]

```

Out[]:=

predictor	mean absolute error
this model, medians at 0 BP	0.146
constant 0.5 for everyone	0.191

Present-day extrapolation versus 1000 Genomes phase-3 allele frequencies (Ensembl). Everything after ~2,000 BP is extrapolation far beyond the ancient data, and the wide intervals say so. The second table is the test that keeps everyone honest: it scores the model's medians against the laziest imaginable null — predicting 0.5 for every population. A spatial diffusion–selection machine that cannot clearly beat a constant has no present-day predictive content, whichever way the comparison lands in the build you are reading; the model is calibrated for the ancient window it was fitted to, and that is the only window it should be quoted for.

Taken together, the verdict is split, and saying so is the point of the section. **Timing lands inside the published envelope** — between Itan et al.'s mode and the imputed-genome rise date — and **selection strength is order-of-magnitude consistent** with all four published anchors. **Location is not identified**, and to the extent the posterior concentrates anywhere under the current data and error model, it leans toward the Atlantic side of Europe rather than the Carpathian Basin — the

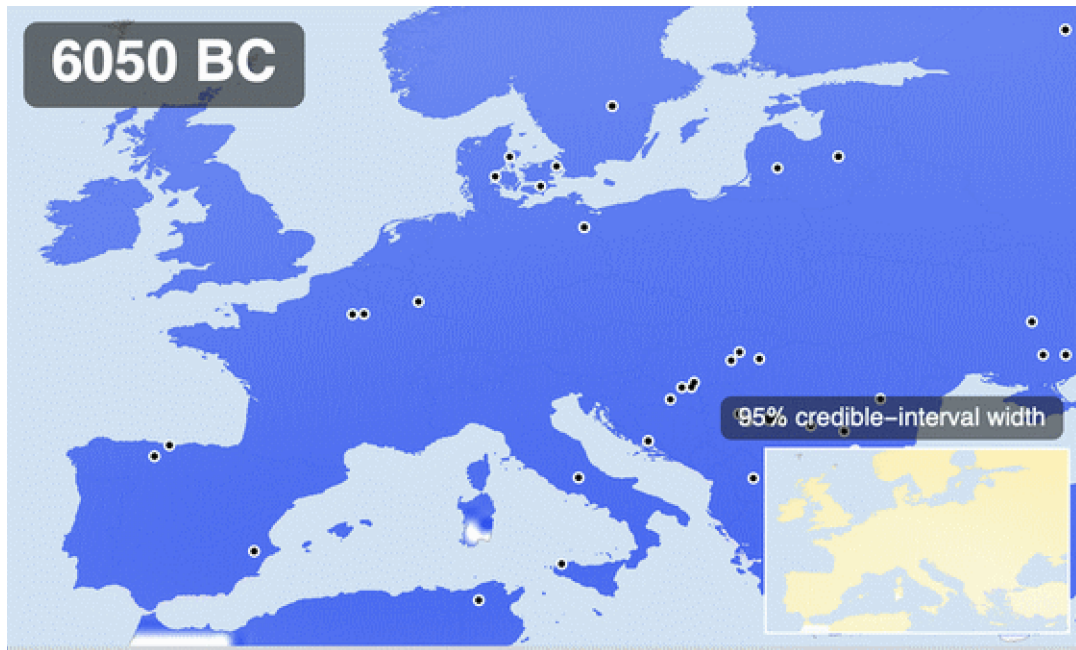
opposite shore from the textbook answer. Failing to exclude the central-European origin is not support for it: this analysis should not be cited as ancient-DNA confirmation of the Carpathian origin, and whether the westward lean is signal or residual artefact (sampling density, the grid's geometry, the absence of drift) is exactly what the assumption-sensitivity experiments in the repository are for.

It is worth being precise about *why* location fails while timing succeeds, because the reason is in the data rather than the code. Of the 636 called alleles older than 6,000 BP in this dataset, **exactly one** is derived — a single heterozygote from Bulgaria at about 6,560 BP. Timing is identified because the whole European frequency curve turns upward at a well-determined moment, and thousands of alleles constrain that turn. Location has essentially one informative carrier plus a field of zeros, and a zero is only weak evidence of absence when sampling is as spatially uneven as this. No amount of extra sequencing fixes that directly, because the allele genuinely was near zero frequency then; the fix has to come from a statistic that reads the geometry of the spread rather than the roster of early carriers. Fitting the *arrival time* as a function of distance from a candidate source — the classical wave-of-advance signal, where a front under Fisher–KPP dynamics advances at $c = 2\sqrt{Ds}$ — uses the thousands of *later* samples to locate the apex of the cone, and is the obvious next step for this project.

12. The animation, assembled

The hero animation at the top of the post is the whole pipeline in one artifact: 81 time-interpolated frames of the kriged posterior mean, the moving window of ancient samples, a progress bar, and — non-negotiably — the credible-interval inset, so that even a viewer who reads nothing else cannot mistake the smooth colours for certainty. Because every frame is raster-composited over a single cached base map, the full render takes under a minute:

```
In[ ]:= hero = ExportHeroAnimation[Directory[], samples, grid, draws];
Out[ ]:=
```



The hero time-lapse, re-rendered by the call above from the posterior draws of this session. MP4 and GIF land in figures/generated/.

13. What this is not

This is a calibrated caricature, and the caveats are part of the result. The regional logistic layer reproduces the published four-region picture qualitatively, not parameter-for-parameter. The simulator is deterministic — no genetic drift — which flatters the posterior's tightness; demography (the steppe migrations!) is implicit; Ireland is unreachable in the adjacency; radiocarbon uncertainty enters the bins as point dates; the final ESS of 16.0 keeps every interval approximate; and everything after ~2,000 BP extrapolates beyond the data. Next steps, in rough order of value: a formal model comparison between the uniform-start and point-source variants (§10), a Wright-Fisher stochastic core, sample-level ancestry-aware likelihoods, MCMC rejuvenation inside the SMC, and calibrated-date uncertainty propagated end-to-end.

14. Reproducibility

Everything — data retrieval with checksums, processing with provenance, fits, SMC, validation, sensitivity, every figure, the hero animation, and this notebook itself — regenerates from the repository:

```
git clone https://github.com/mthiel74/ancient-dna-lactase-spatial-wolfram
wolframscript -file scripts/run_pipeline.wls --particles 400 --generations 5
wolframscript -file scripts/run_tests.wls
wolframscript -file community/build_notebook.wls
```

The package `src/LactasePersistenceSpatial.wl` (~2,000 lines) carries the entire workflow behind the one-liners in this post; 24 `VerificationTests` cover the parsers, the simulator, the summary statistics, and the SMC machinery. Repository: [mthiel74/ancient-dna-lactase-spatial-wolfram](https://github.com/mthiel74/ancient-dna-lactase-spatial-wolfram).

15. References

Evershed, R. P., et al. (2022). Dairying, diseases and the evolution of lactase persistence in Europe. *Nature* 608, 336–345. [nature.com/articles/s41586-022-05010-7](https://www.nature.com/articles/s41586-022-05010-7)

Burger, J., et al. (2020). Low prevalence of lactase persistence in Bronze Age Europe indicates ongoing strong selection over the last 3,000 years. *Current Biology* 30(21), 4307–4315. [cell.com/current-biology](https://www.cell.com/current-biology)

Bersaglieri, T., et al. (2004). Genetic signatures of strong recent positive selection at the lactase gene. *American Journal of Human Genetics* 74(6), 1111–1120.

Irving-Pease, E. K., et al. (2024). The selection landscape and genetic legacy of ancient Eurasians. *Nature* 625, 312–320. [nature.com/articles/s41586-023-06705-1](https://www.nature.com/articles/s41586-023-06705-1)

Itan, Y., Powell, A., Beaumont, M. A., Burger, J., Thomas, M. G. (2009). The origins of lactase persistence in Europe. *PLoS Computational Biology* 5(8), e1000491 (CC-BY; Figure 3 reproduced in §11).

The 1000 Genomes Project Consortium (2015). A global reference for human genetic variation. *Nature* 526, 68–74. rs4988235 population frequencies via Ensembl.

GLAD LP Ancient Genotypes 2022 workbook (UCL), derived from the Allen Ancient DNA Resource v44.3.

Analysis, figures, and notebook generated with Wolfram Language 15.0.1 on September 1, 2026.